“Smart City Waste Monitoring System”

A PROJECT REPORT

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In fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

INFORMATION TECHNOLOGY

LDRP Institute of Technology and Research, Gandhinagar

Kadi Sarva Vishwavidyalaya

April-2025-26

LDRP INSTITUTE OF TECHNOLOGY AND RESEARCH GANDHINAGAR

IT Department

CERTIFICATE

This is to certify that the Project Work entitled **"Smart City Waste Monitoring System"** has been carried out by **PATEL DIYA (23BEIT30103)** under my guidance in fulfilment of the degree of Bachelor of Engineering in Information Technology Semester-6 of Kadi Sarva Vishwavidyalaya during the academic year 2025-2026.

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Presentation-I for Project-I

1. Name & Signature of Internal Guide	
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Presentation-II for Project-I

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3. Name & Signature of Panel Members	

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who contributed to the successful completion of my project titled *"Smart City Waste Monitoring System."* First and foremost, I extend my respectful thanks to the Head of the Department, **Dr. Mehul Barot**, for his valuable guidance, encouragement, and support throughout the course of this project. I am also deeply grateful to my project guide, **Prof. Rashmika Patel**, for her continuous guidance, constructive suggestions, and constant motivation, which greatly contributed to the successful completion of this work. I would also like to thank the **IT Department, LDRP-ITR**, for providing the necessary facilities, infrastructure, and resources required to carry out this project successfully.

Finally, I acknowledge with gratitude the contributions of various researchers and developers in the field of Smart Systems, whose work served as a valuable reference and source of inspiration for this project. This project has been a highly enriching learning experience, and I am truly thankful for the opportunity to explore, design, and implement the Smart City Waste Monitoring System.

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ABSTRACT

The **Smart City Waste Monitoring System** is a web-based application designed to streamline the process of waste management and complaint resolution in urban areas. Traditional methods of reporting waste disposal issues to municipal authorities are often cumbersome and inefficient, leading to delays in addressing environmental hazards. This project aims to bridge the gap between citizens and municipal bodies by providing a digital platform for efficient grievance redressal.

The system allows citizens to register complaints regarding uncollected garbage or waste mismanagement in their locality directly through an intuitive user interface. Key features include the ability to submit complaints with location details, track the status of reported issues (e.g., pending or resolved), and access educational resources on waste recycling and segregation.

On the administrative side, the platform empowers municipal authorities to monitor incoming complaints, acknowledge reports, and deploy resources effectively to ensure timely resolution. Developed using **PHP, HTML, CSS, and Bootstrap**, the application ensures a responsive and user-friendly experience across devices. By automating the complaint handling process and promoting transparency, the Smart City Waste Monitoring System contributes to a cleaner, healthier, and more sustainable urban environment.

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1 INTRODUCTION

1.1. INTRODUCTION

1.2. AIMS AND OBJECTIVE OF THE WORK

1.3. BRIEF LITERATURE REVIEW

1.4. PROBLEM DEFINITION

1.5. PLAN OF THEIR WORK

1.1. Introduction

The **Smart City Waste Monitoring System** is a responsive web application designed to modernize and streamline municipal waste management by fostering direct collaboration between citizens and local authorities. In rapidly growing urban environments, the traditional methods of reporting waste disposal issues—often involving physical visits to municipal offices or lengthy bureaucratic processes—are inefficient and discourage public participation. This project addresses these challenges by providing a centralized digital platform that makes waste monitoring and grievance redressal accessible, transparent, and effective.

At its core, the system empowers citizens to play an active role in maintaining cleanliness in their communities. Users can easily register complaints regarding uncollected garbage, hazardous waste, or irregular sanitation services in their locality. The platform not only facilitates the submission of these reports but also ensures accountability by allowing users to track the status of their complaints, verifying whether the reported issues have been acknowledged and resolved by the authorities.

For municipal administrators, the system acts as a powerful management tool. It aggregates complaints from various locations, enabling authorities to identify problem areas, allocate resources efficiently, and monitor the performance of sanitation crews. By automating the flow of information from the public to the administration, the system significantly reduces response times and operational inefficiencies.

Additionally, the project aims to educate the public on sustainable practices. It features dedicated sections on waste segregation, recycling guidelines, and the proper disposal of medical and electronic waste, promoting a culture of environmental responsibility. By integrating technology with civic engagement, the Smart City Waste Monitoring System strives to create cleaner, healthier, and smarter cities for the future.

1.2. Aims and Objective of the work

1.2.1. Aims of "Smart City Waste Monitoring System"

The Smart City Waste Monitoring System is a web application designed to streamline waste management by connecting citizens directly with their municipalities.

Based on the project's documentation, its primary aims are:

- **Digital Complaint System:** To provide a platform for citizens to easily lodge complaints about waste or garbage issues in their locality.
- **Transparency & Tracking:** To allow users to track the status of their complaints and verify whether the reported issues have been resolved.
- **Process Automation:** To replace cumbersome manual processes with an automated system that redirects complaints to the appropriate municipality admins for acknowledgement and action.

- **Education:** To provide citizens with ideas and information regarding waste recycling.

In short, it aims to make the process of reporting and managing waste issues easier, faster, and more transparent for both citizens and municipal authorities.

1.2.2. Objectives of "Smart City Waste Monitoring System"

The Smart City Waste Monitoring System enables citizens to easily report garbage issues for timely resolution by municipal authorities. It streamlines waste management operations while educating the public on proper disposal practices to ensure a cleaner and healthier city.

- **Digitize Complaints:** Replace manual waste reporting with an efficient web-based system.
- **Ensure Accountability:** Enable citizens to track the status of their reported issues in real-time.
- **Streamline Operations:** Provide authorities with a centralized dashboard to manage and resolve garbage alerts.
- **Educate Citizens:** Disseminate clear guidelines on proper waste segregation and recycling.
- **Improve Hygiene:** Accelerate waste collection to maintain cleaner, healthier urban environments.

These objectives together aim to make **Smart City Waste Monitoring System** an efficient, reliable, and intelligent platform for managing digital Waste Management effectively.

1.3. Brief Literature Review

The development of the Smart City Waste Monitoring System is grounded in existing research and guidelines regarding waste management challenges, particularly in the context of healthcare and municipal oversight.

1.3.1. National Health Care Waste Management Guidelines Research indicates a critical need for strict adherence to waste management protocols. According to the **National Health Care Waste Management Guidelines** prepared by the **National Health Research Council (NHRC)**, healthcare facilities are mandated to segregate waste into infectious and non-infectious categories. However, implementation remains a challenge due to budget constraints and lack of technical orientation for waste handlers.

1.3.2. Institutional Challenges & Reports The **Department of Health Services' Annual Health Report (2000/2001)** highlighted the extensive network of healthcare institutions in Nepal (74 hospitals, 172 PHCs, etc.), all of which generate significant waste. Current literature, including statements from hospital administration (e.g., **Bir Hospital**), suggests that infrastructure damage (such as from the 2015 earthquake) and lack of space have forced many institutions to revert to traditional, less safe sterilization methods, bypassing modern guidelines.

1.3.3. Municipal Waste Composition Understanding the composition of waste is vital for effective monitoring. Drawing from similar studies, such as ″Management of Municipal Solid Wastes: A Case Study in Limpopo Province, South Africa,″ typical municipal waste is composed of approximately:

- **Plastics (35%)**
- **Paper and Glass (25%)**
- **Food Waste (25%)**
- **Garden Waste (15%)**

This data underscores the importance of a system that not only monitors waste levels but also encourages segregation and recycling, which is a core objective of this project.

1.4. Problem Definition

The current state of waste management in the city faces several critical challenges that this project seeks to address:

- **Inefficient Complaint Mechanism**: The existing process for citizens to report waste or garbage accumulation is a "hefty process," often requiring physical visits or manual communication with the municipality. This barrier discourages citizens from reporting problems, leading to prolonged waste accumulation.
- **Lack of Transparency and Feedback**: Citizens who do report issues often have no way to track the status of their complaints. There is a "disconnect" where users do not know if their report has been acknowledged or if the work has been completed.
- **Public Health and Environmental Risks**: As highlighted by the National Health Care Waste Management Guidelines, strict adherence to waste segregation is often ignored due to lack of resources (budget and infrastructure) and monitoring. This leads to haphazard management—such as mixing medical waste with municipal waste—which poses significant risks to public health and the environment.
- **Operational Gaps**: There is a lack of a centralized digital platform to bridge the gap between the "concerns of general citizens" and their medical/municipal authorities, resulting in delayed responses and compromised city hygiene.

1.5. Plan of Their Work

To ensure the successful development and deployment of the Smart City Waste Monitoring System, the project work is divided into the following strategic phases:

Phase 1: Requirement Analysis & Feasibility

- **Objective**: Identify the core problems in current waste management and define project scope.

- **Activities**:
    - Studying national waste management guidelines (NHRC) and reviewing existing literature.
    - Defining user roles: **General Citizens** (complainants) and **Municipality Admins** (supervisors).
    - Determining technical requirements: Web-based platform accessible via handheld devices.

Phase 2: System Design

- **Objective**: Architect the database and user interface.
- **Activities**:
    - **Database Design**: Creating the wms.sql schema to store admin credentials, user data, and complaint records (garbageinfo table).
- **UI/UX Design**: Designing responsive front-end interfaces using **HTML5, CSS3, and Bootstrap** to ensure compatibility across mobile and desktop devices.
- **Workflow Mapping**: Designing the flow for complaint submission, email verification (phpGmailSMTP), and status updates.

Phase 3: Implementation (Development)

- **Objective**: Coding the functional modules.
- **Activities**:
    - **Authentication Module**: Developing secure Login/Signup pages (login-user.php, signup-user.php) with OTP verification.
- **Complaint Module**: Building the trash.php interface for users to submit waste reports.
- **Admin Dashboard**: Creating the control panel (adminlogin) for officials to view, verify, and update the status of complaints.
- **Backend Integration**: connecting frontend forms to the **MySQL** database using **PHP**.

Phase 4: Testing & Validation

- **Objective**: Ensure system reliability and bug fixing.
- **Activities**:
    - **Unit Testing**: Verifying individual modules like connection establishment (

connection.php) and email delivery.

- **Integration Testing**: Checking the redirection flow from user login -> complaint filing -> admin acknowledgement.
- **User Acceptance Testing (UAT)**: Validating that the "status update" feature correctly reflects changes on the user's dashboard.

Phase 5: Deployment & Maintenance

- **Objective**: Go-live and future improvements.
- **Activities**:
    - Deploying the application to a local server (initial phase) or public web server.
    - Monitoring system performance and gathering user feedback for future features like real-time GPS tracking of waste trucks.

2 TECHNOLOGY AND LITERATURE REVIEW

2.1. TECHNOLOGY

2.2. LITERATURE REVIEW

2.1. Technologies

2.1.1. Frontend Technologies (Client-Side)

- **HTML5**: Used for the semantic structure of the application (headers, sections, footers).
- **CSS3**: Custom styling implemented in

  style.css and

  wavestyle.css.

- **Bootstrap**: A front-end framework used for:
  - Responsive design (ensuring compatibility across mobile, tablet, and desktop).
  - Pre-styled components (navbars, buttons, forms, grids).
- **JavaScript & jQuery**:
  - Used for client-side logic and DOM manipulation.
  - **Form Validation**: validate.js ensures user inputs are correct before submission.
  - **Dynamic UI**: Controls animations and interactive elements.
- **UI Libraries & Plugins**:
  - **AOS (Animate On Scroll)**: Adds fade/zoom animations as the user scrolls.
  - **GLightbox**: A lightbox script for displaying images (gallery/waste reports) in a popup.
  - **Swiper**: Used for touch-enabled sliders/carousels (e.g., testimonials or clients).
  - **Waypoints**: Triggers functions when scrolling to specific elements.
- **Icons**:
  - **FontAwesome**, **Boxicons**, **RemixIcon**: Used for vector icons (trash cans, user profiles, social media logos).
- **Fonts**:
  - **Google Fonts**: Uses 'Open Sans', 'Jost', and 'Poppins' for modern typography

2.1.2. Backend Technologies (Server-Side)

- **PHP (Hypertext Preprocessor)**: The primary server-side scripting language.

- **Core Logic**: Handles user requests, processes forms, and manages sessions.
- **File Handling**: Manages the upload of waste images to the server.
- **Controller Pattern**: controllerUserData.php acts as a central controller handling logic for login, signup, and profile management.
- **Email Services**:
  - **PHPMailer (via phpGmailSMTP)**: Integration with SMTP to send transactional emails (OTP verification, password resets).

2.1.3. Database System

- **MySQL (MariaDB)**: A relational database management system.
  - **Storage Engine**: Uses **InnoDB** for reliable transaction processing.
  - **Structure**: Organizing data into related tables:
    - usertable: Stores citizen credentials.
    - adminlogin / adminlogin_tbl: Stores administrator credentials.
    - garbageinfo: Stores report details (location, waste type, image, status).
    - contact: Stores user feedback/inquiries.

2.1.4. Security Implementation

- **Password Hashing**: Uses **Bcrypt** (via PHP's password_hash function) to secure user passwords in the database (evident from the $2y$10$ prefix in the SQL dump).
- **Input Validation**: Protects against basic injections by sanitizing inputs.
- **OTP Verification**: Implements a One-Time Password mechanism (user-otp.php) for verifying user email addresses during signup.
- **Session Management**: Uses PHP Sessions to manage user login states and separate User/Admin access levels.

2.2. Literature Review

2.2.1. Contextual Background

- Urbanization Challenge: Rapid urbanization in cities (specifically referenced in the project context of Nepal, e.g., Kathmandu, Bhaktapur) has outpaced traditional waste management infrastructure.
- Public Health Impact: The project highlights that shortcomings in hospital staff and management regarding waste disposal can have adverse effects on public health.
- Current Infrastructure: The system operates within a context where government-run hospitals (74+) and primary health care units exist but struggle with implementation.

2.2.2. Regulatory Framework & Guidelines

- National Guidelines: The project explicitly references the "National Health Care Waste Management Guidelines" prepared by the National Health Research Council (NHRC).
- Mandatory Compliance: All healthcare facilities are required to follow these guidelines, but actual implementation remains inconsistent.
- Waste Categorization: The literature identifies two critical categories of waste that must be managed differently:
  - Infectious Waste: Human/animal biological waste, pathogen-contaminated items.

2.2.3. Problem Statement & Gaps

- Implementation Gap: The project notes that despite guidelines, many hospitals fail to nts Non-infectious Waste: Toxic chemicals, radioactive materials, and general .wasteto detailed issues like the 2015 earthquake damaging sterilization facilities (autoclaves) and lack of space for new installations.

- Private vs. Public Sector: A distinction is drawn where private hospitals often manage waste more pfollow them due to:
  - Lack of budget.
  - Lack of staff orientation/training.
  - Flimsy monitoring from the government.
- Infrastructure Deficits: Qualitative data (citing Dr. Kedar Century, Executive Director at Bir Hospital) poioorly than government ones due to a lack of regular monitoring, often burning or burying hazardous waste illicitly

2.2.4. The Role of Digital Intervention (The Project's Contribution)

- From Passive to Active: Shifts waste management from a passive municipal service to an active, citizen-driven process.
- Crowdsourced Monitoring: Utilizes the "Smart City" concept where citizens act as sensors throughout the city, reporting waste accumulation in real-time.
- Transparency & Accountability: The digital nature of the system (logging complaints in a database) prevents reports from being ignored, unlike traditional verbal or paper-based complaints.
- Education as a Tool: The platform functions not just as a reporting tool but as a knowledge hub, educating users on proper waste segregation (Wet, Dry, E-waste, Biomedical) to solve the problem at the source.

3 SYSTEM REQUIREMENTS STUDY

3.1. USER CHARACTERISTICS

3.2. HARDWARE AND SOFTWARE REQUIREMENTS

3.3. ASSUMPTIONS AND DEPENDENCIES

3.1. User Characteristics

The **Smart City Waste Monitoring System** is designed for two distinct user groups, each with specific roles and technical aptitudes.

3.1.1. General Citizens (Users)

- **Role**: The primary data contributors who report waste issues.
- **Characteristics**:
  - **Demographics**: Residents of the city, varying widely in age and educational background.
  - **Technical Expertise**: Low to Medium. The system is designed with a simple UI (index.html) to accommodate users with basic smartphone knowledge.
  - **Frequency of Use**: Sporadic (whenever they encounter waste).
- **Key Privileges**:
  - Register/Login signup-user.php).
- Upload images of waste (garbageinfo table).
- View status of their reported complaints.

3.1.2. Municipal Administrators (Admin)

- **Role**: The managers responsible for viewing reports and deploying cleaning crews.
- **Characteristics**:
  - **Demographics**: Municipal employees or waste management supervisors.
  - **Technical Expertise**: Medium. Required to navigate the backend dashboard (adminlogin/ folder).
  - **Frequency of Use**: Daily (monitoring the dashboard).
- **Key Privileges**:
  - Access to the Admin Dashboard.
  - Authority to update complaint status (e.g., set to "Completed").
  - Manage user accounts and view system-wide statistics.

3.2. Hardware and Software Requirements

3.2.1. Hardware Requirements

Since this is a web-based application, the hardware requirements are minimal for the end-user but specific for the hosting environment.

Client-Side (User/Admin Device)

- **Device**: Smartphone, Tablet, or PC/Laptop.
- **Processor**: 1 GHz or faster.
- **RAM**: Minimum 512 MB (Standard for running modern browsers).
- **Internet Connection**: 3G/4G/5G or Wi-Fi is mandatory for uploading images.

Server-Side (Hosting Environment)

- **Processor**: Intel Core i3 or equivalent virtual private server (VPS) vCPU.
- **RAM**: Minimum 2 GB (Recommended for running Apache + MySQL).
- **Storage**: 10 GB HDD/SSD (To store the OS, Database, and uploaded user images).

3.2.2. Software Requirements

Development Environment (What was used to build it)

- **Operating System**: Windows 10/11 (Development) or Linux (Production).
- **Code Editor**: Visual Studio Code / Sublime Text.
- **Local Server**: XAMPP / WAMP (Control Panel v3.2 or higher).

Technology Stack (Runtime Environment)

- **Web Server**: Apache HTTP Server (v2.4+).
- **Database**: MySQL (v5.7 or MariaDB 10.4+).
- **Server-Side Language**: PHP (v7.4 or higher recommended for security).
- **Client-Side**: Any modern Web Browser (Chrome, Firefox, Edge, Safari).

3.3. Assumptions and Dependencies

3.3.1. Assumptions

- **Internet Availability**: It is assumed that the user (citizen) has an active internet connection at the time of reporting the waste. Offline syncing is not currently supported.

- **GPS Functionality**: It is assumed that the user is reporting the waste from the location where it is present, or provides an accurate description in the locationdescription field (garbageinfo table).

- **Image Authenticity**: The system assumes that images uploaded by users are genuine and recent, though the "Two-Factor Authentication" (user-otp.php) helps mitigate spam.

- **Browser Compatibility**: It is assumed users have JavaScript enabled in their browsers for the frontend (bootstrap, aos) to function key features.

3.3.2. Dependencies

- **SMTP Server**: The system relies on **Gmail SMTP** (phpGmailSMTP) for sending emails (OTPs, notifications). Use of this feature depends on Google's "App Password" or "Less Secure App" settings being configured correctly.

- **Hosting Provider**: The application depends on a hosting provider (local localhost or remote) that supports PHP and MySQL.

- **Bootstrap CDN**: The application UI (assets/vendor/bootstrap) depends on the availability of Bootstrap files, which are currently included locally (assets/), removing the strict dependency on an external CDN

4 SYSTEM DIAGRAMS

4.1. ENTITY RELATIONSHIP DIAGRAM

4.2. CLASS DIAGRAM

4.3. USE CASE DIAGRAM

4.4. SEQUENCE DIAGRAM

4.5. ACTIVITY DIAGRAM

4.6. DATA FLOW DIAGRAM

4.1. Entity-Relationship Diagram

An Entity-Relationship Diagram (ERD) is a visual representation used in database design to illustrate the structure and relationships between various entities or objects within a system. It employs entities (representing real-world objects or concepts), attributes (properties of these entities), and relationships (connections between entities) to depict how data is organized and interrelated in a database. ERDs are invaluable tools for database designers and developers as they provide a clear and concise overview of the data model, aiding in the design, organization, and understanding of complex databases, ultimately facilitating efficient data management and retrieval.

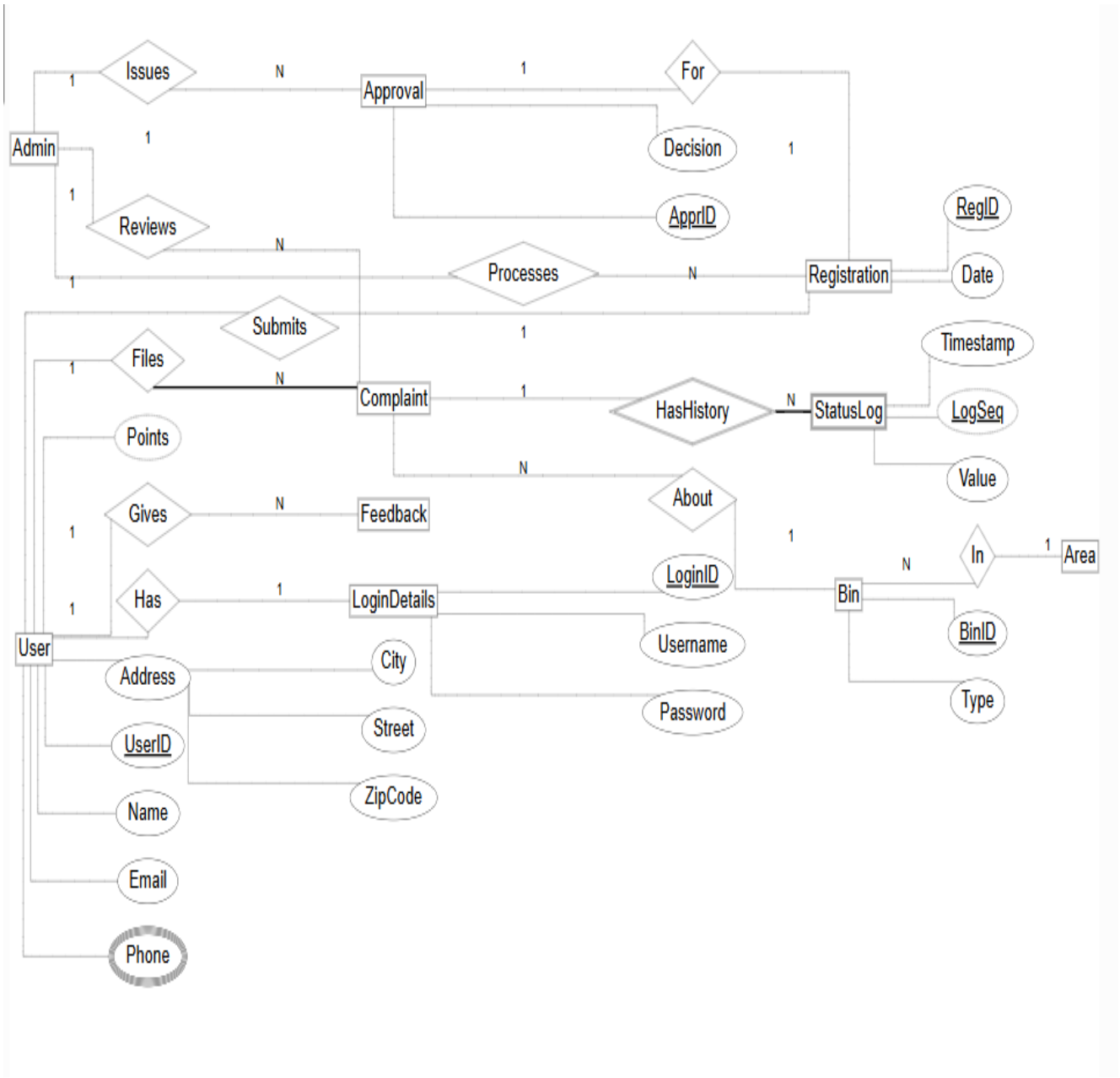

Fig 4.1: Entity-Relationship Diagram

4.2. Class Diagram

A class diagram is a visual representation in UML (Unified Modeling Language) that depicts the structure and relationships of classes within a software system or application. It serves as a blueprint for designing and understanding the software's object-oriented architecture. In a class diagram, classes are depicted as rectangles with attributes and methods, showing how they interact with one another through associations, aggregations, generalizations, and dependencies. This diagram provides a high-level overview of the system's components, helping software developers and stakeholders to conceptualize, communicate, and plan the software's design and functionality effectively.

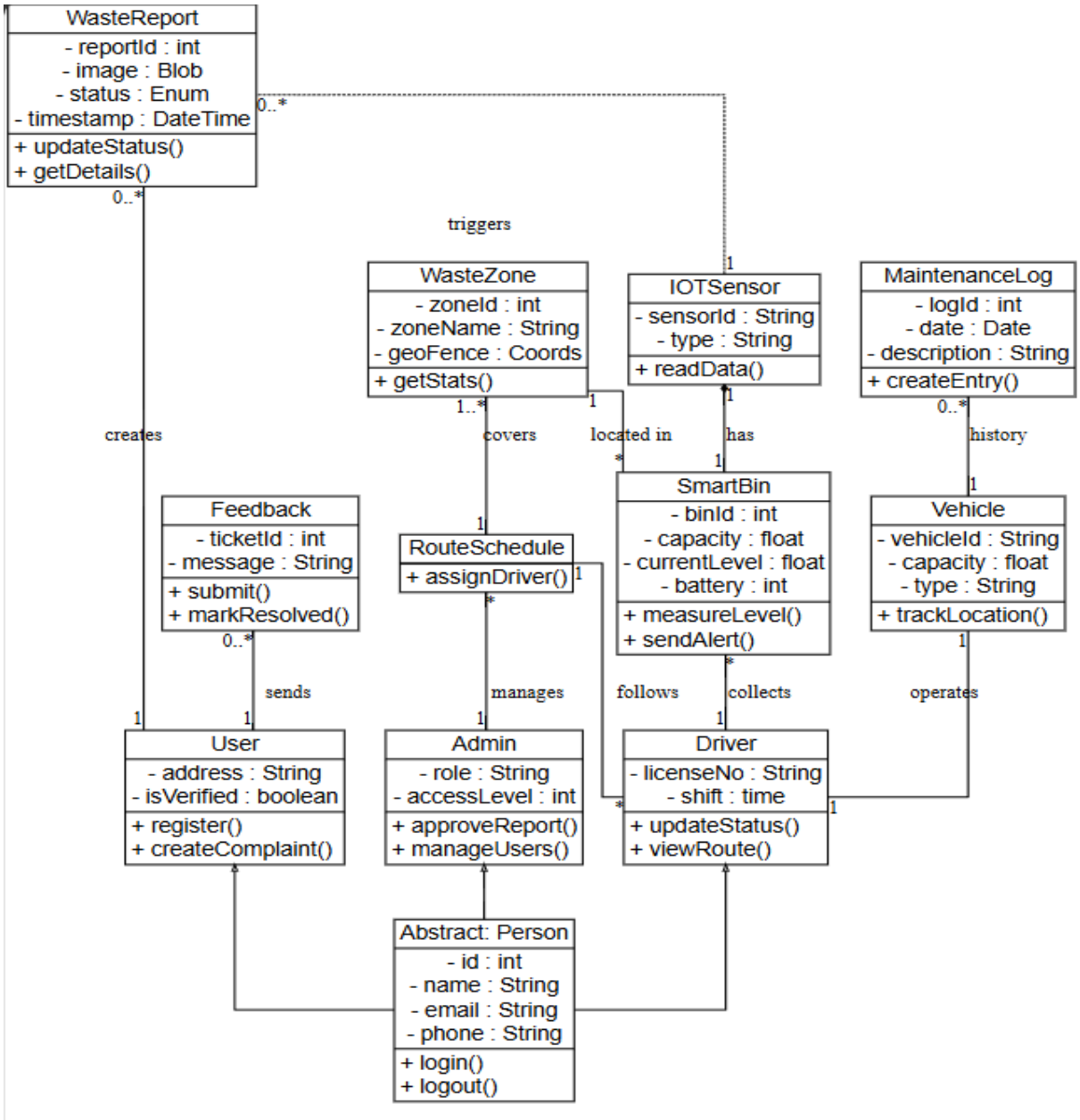

Fig 4.2: Class Diagram

4.3. Use Case Diagram

A use case diagram in software engineering is a visual representation that depicts the interactions between various actors (which can be individuals, other systems, or entities)and a system or software application. It serves to illustrate the different ways users or external entitie scaninteract with the system,show casing the system's functionality fromauser' s perspective. Each use case represents a specific functionality or action that a user oractor can perform within the system. Arrows or lines connecting actors to use cases illustrate the relationships and interactions between them. Use case diagrams are valuable for clarifying system requirements, documenting user goals, and providing a high-level over view of the system's behavior without delving into technical details.

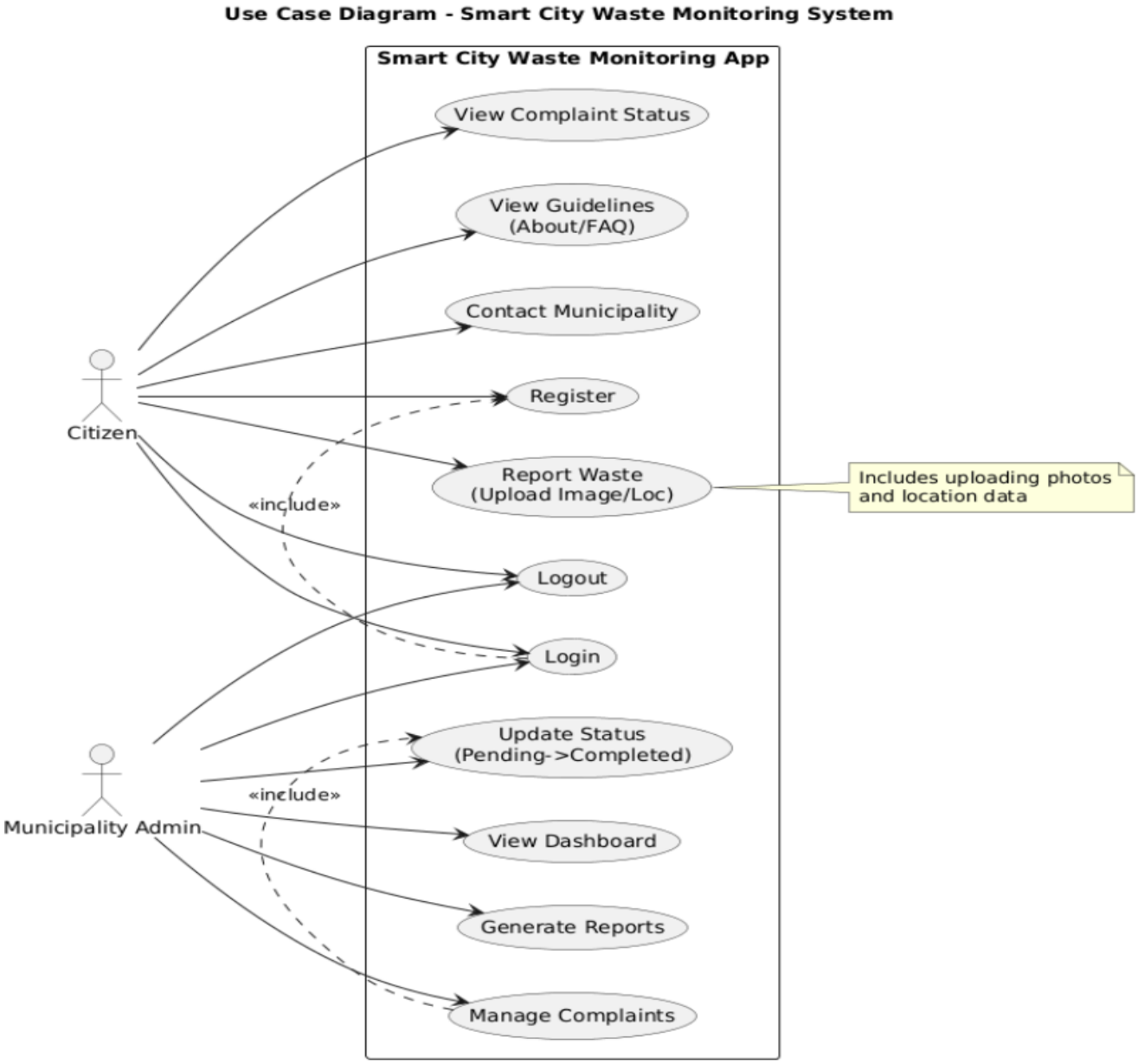

Fig 4.3: Use Case Diagram

4.4. Sequence Diagram

A sequence diagram consists of a group of objects that are represented by lifelines, and the messages that they exchange over time during the interaction. A sequence diagram shows the sequence of messages passed between objects. Sequence diagrams can also show the control structures between objects.

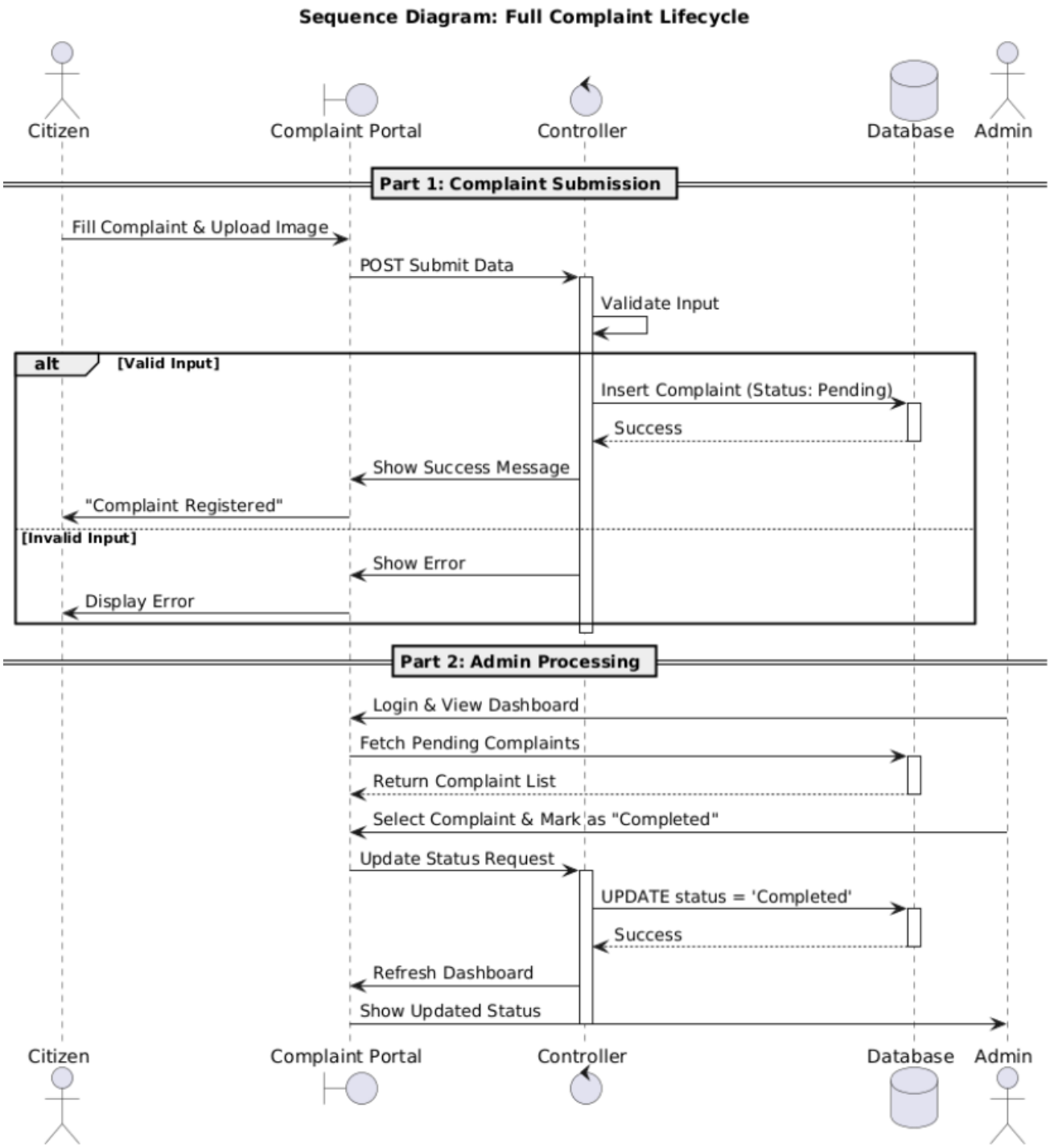

Fig 4.4: Sequence Diagram

4.5. Activity Diagram

Activity diagram is another important behavioral diagram in UML diagram to describe dynamic aspects of the system. Activity diagram is essentially an advanced version of flow chart that modeling the flow from one activity to other activity.

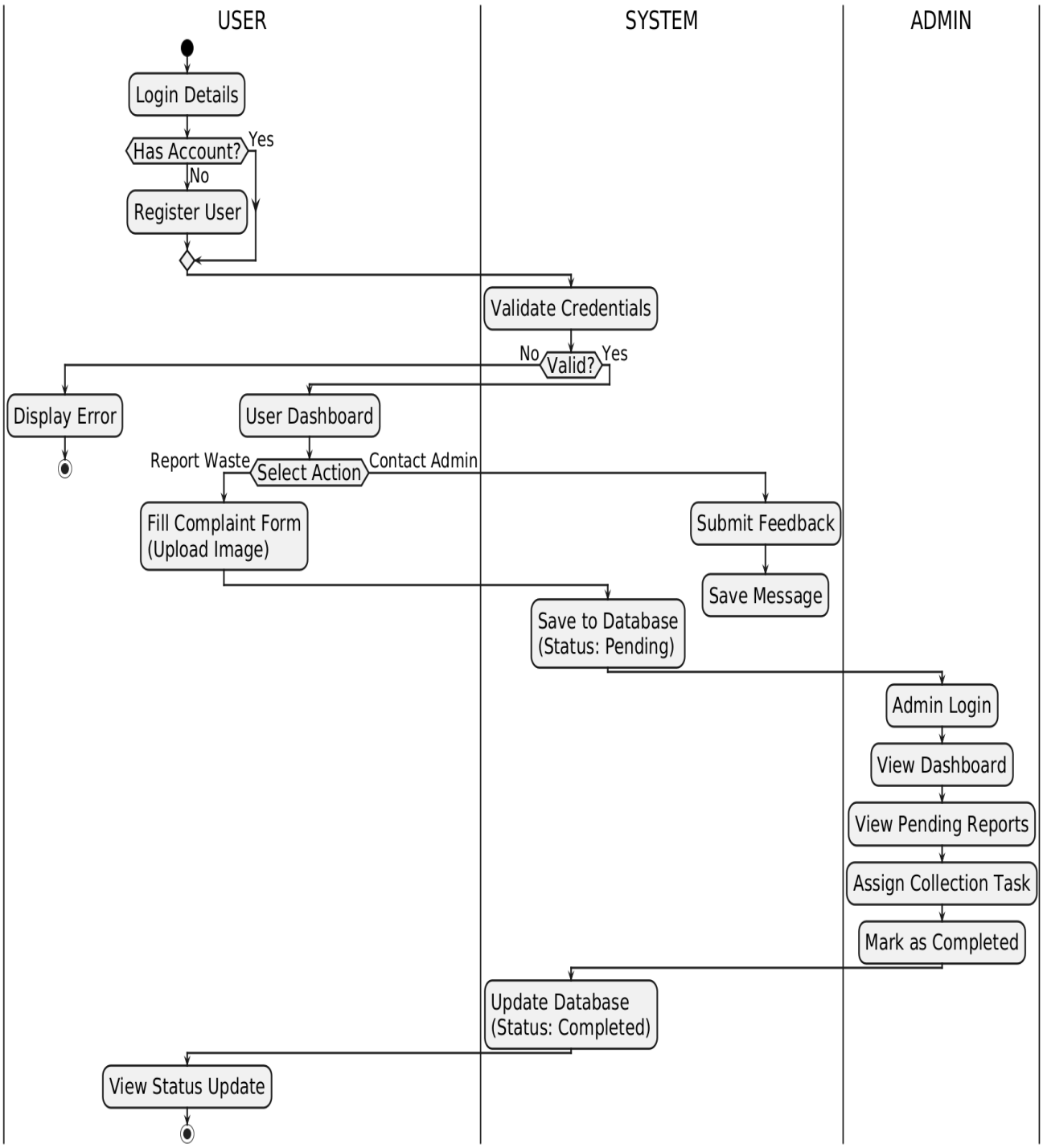

Fig 4.5: Activity Diagram

4.6. Data flow Diagram

A Data Flow Diagram (DFD) is a visual representation that depicts the flow of data within a system or process. It uses various symbols to illustrate how data moves from input sources through processes to output destinations, showing the transformation and interaction of data along the way. DFDs are valuable tools in systems analysis and design, allowing stakeholders to gain insights into how data is processed and exchanged within a system, highlighting potential bottlenecks, inefficiencies, and areas for improvement. They are commonly used in information systems and software engineering to document and analyze data flow, making them instrumental in understanding and optimizing the functionality and efficiency of complex systems.

4.6.1. Context-Level-0 DFD:

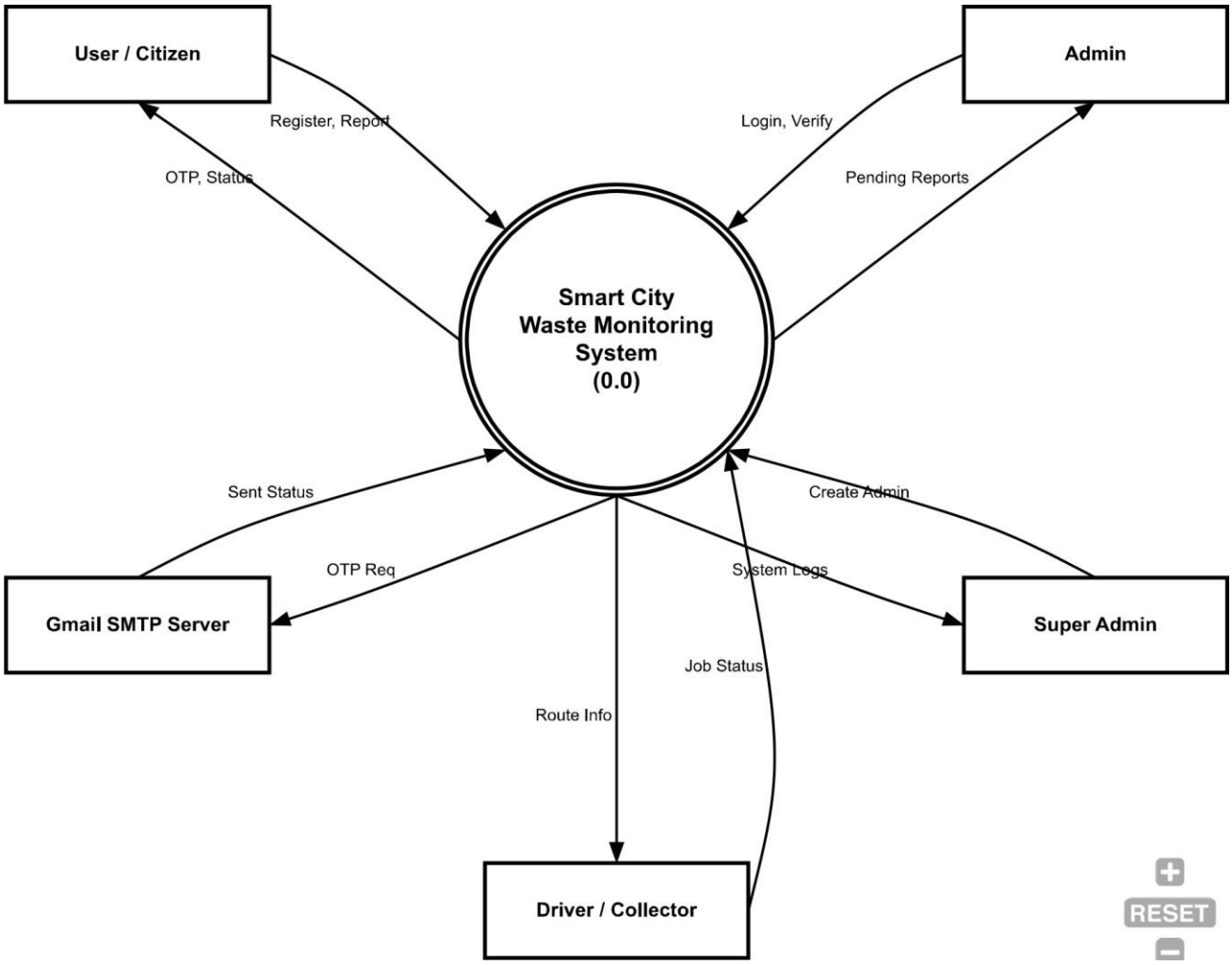

Fig 4.6.1: Context-Level-0 Data Flow Diagram

4.6.2. Context-Level-1 DFD:

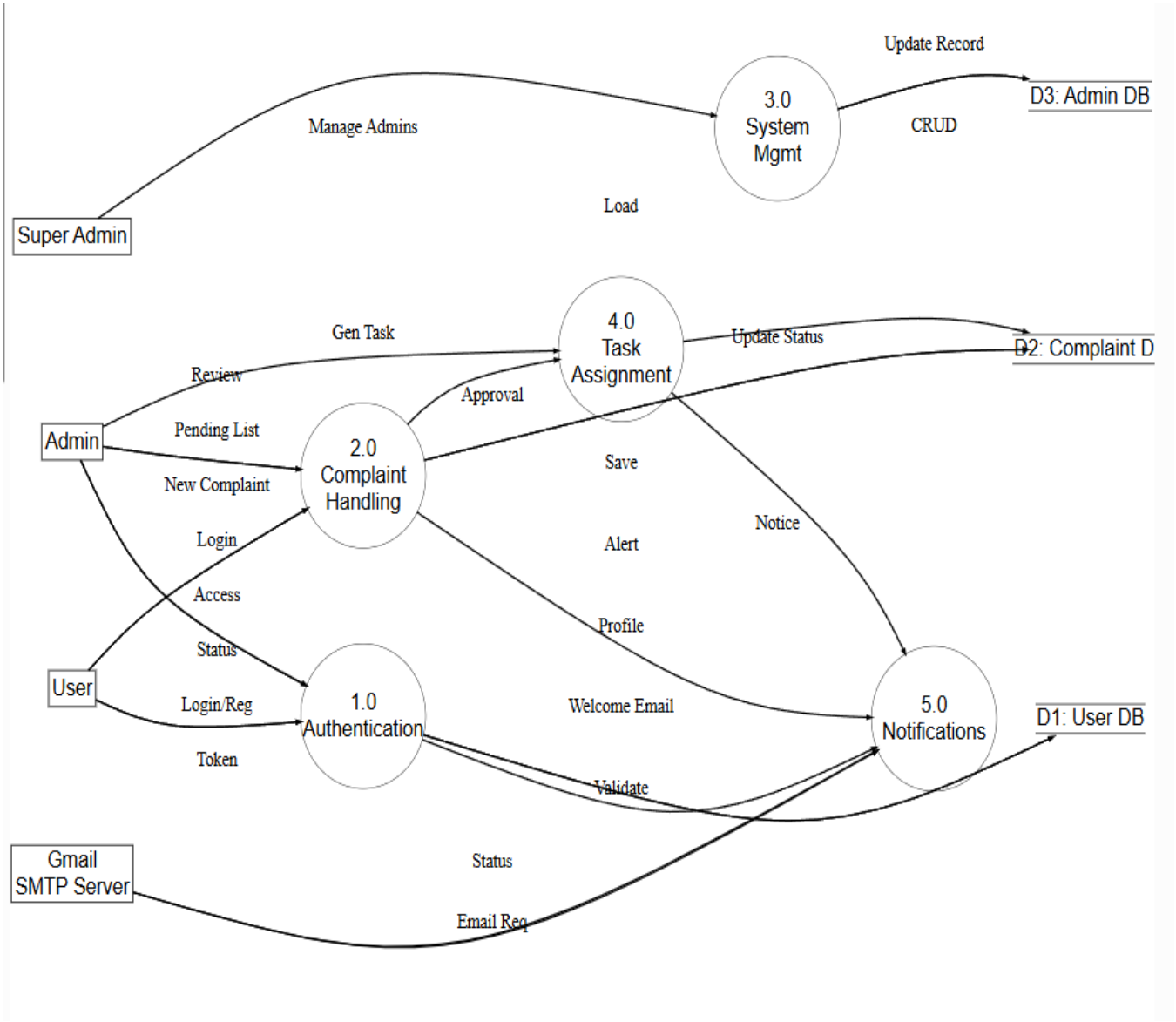

Fig 4.6.2: Context-Level-1 Data Flow Diagram

4.6.3. Context-Level-2 DFD For User :

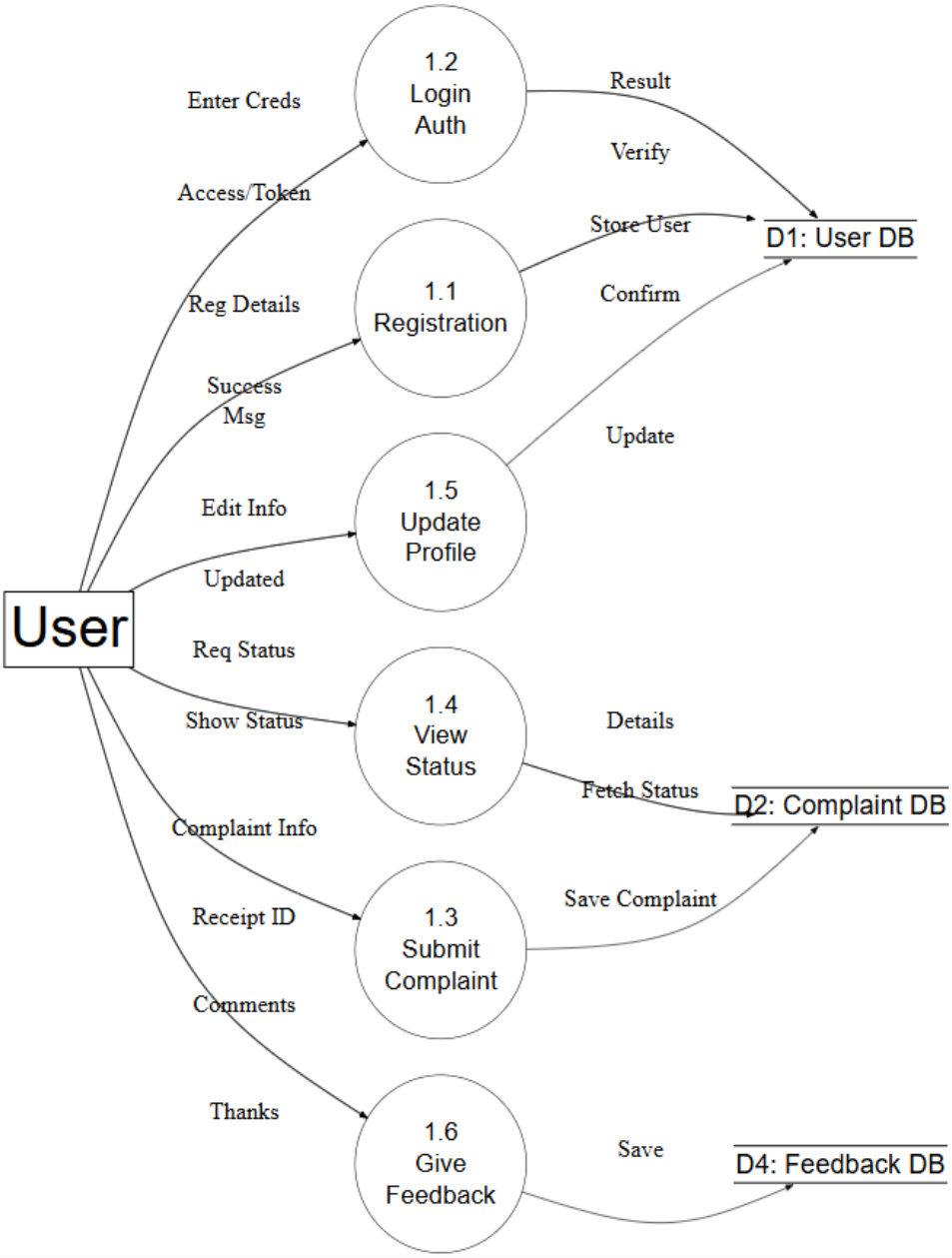

Fig 4.6.3: Context-Level-2 Data Flow Diagram For User

4.6.4. Context-Level-2 DFD For Admin :

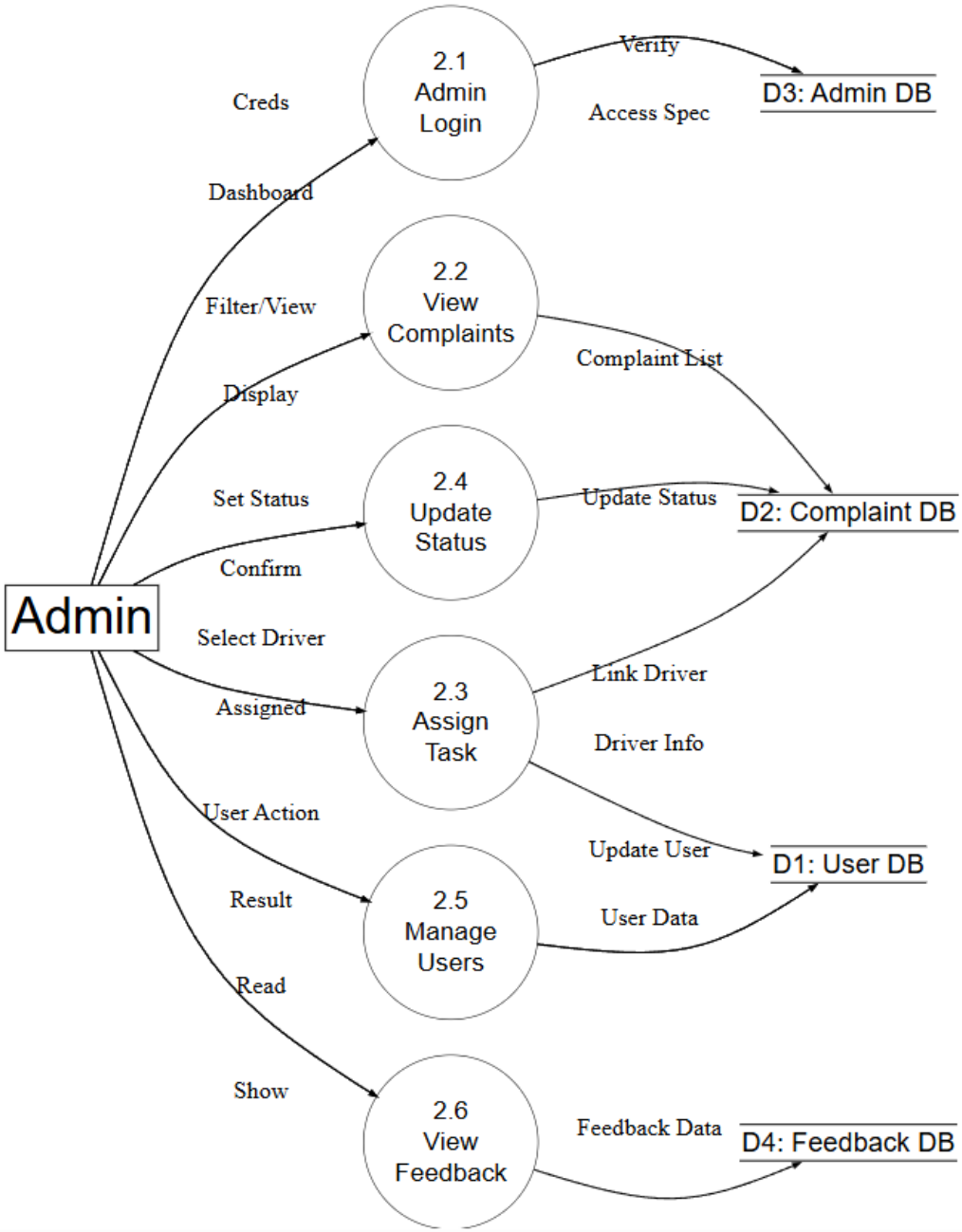

Fig 4.6.4: Context-Level-2 Data Flow Diagram For Admin

5 DATA DICTIONARY

5.1. OVERVIEW

5.2. DATABASE SCHEMA

5.3. DATA INTEGRITY AND NORMALIZATION

5.1. Overview

The **Smart City Waste Monitoring System** utilizes **MySQL**, a relational database management system (RDBMS), to store and manage all application data. The database is named wms and interacts with the PHP backend to perform CRUD (Create, Read, Update, Delete) operations.

5.2. Database Schema

The database consists of **5 tables** designed to handle user authentication, administrative control, waste reporting, and public inquiries.

5.2.1. Table: User table

This table stores the registration details of the citizens (end-users) who report waste.

Column Name	Data Type	Constraints	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each user.
name	VARCHAR(255)	NOT NULL	Full name of the citizen.
email	VARCHAR(255)	NOT NULL	Email address (used for login and OTP).
password	VARCHAR(255)	NOT NULL	Hashed password for security.
code	MEDIUMINT(50)	NOT NULL	OTP code used for verification.
status	TEXT	NOT NULL	Account status (e.g., 'verified').

5.2.2. Table: Garbage info

This is the core transactional table that stores the details of waste reported by citizens.

Column Name	Data Type	Constraints	Description
Id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each complaint.
name	VARCHAR(255)	NOT NULL	Name of the reporter.
mobile	VARCHAR(15)	NULL	Contact number of the reporter.
email	VARCHAR(255)	NOT NULL	Email to link report to user account.
wastetype	VARCHAR(255)	NOT NULL	Category of waste (e.g., Organic, Mixed).
location	VARCHAR(255)	NOT NULL	Area or City name.
locationdescription	VARCHAR(255)	NOT NULL	Detailed landmark or address.
file	VARCHAR(255)	NOT NULL	Filename of the uploaded image proof.
date	VARCHAR(255)	NULL	Timestamp of when the report was submitted.
status	VARCHAR(50)	NULL	Current state (e.g., 'Pending', 'Completed').

5.2.3. Table: Admin login_tbl

This table manages the credentials for municipal administrators who oversee the system.

Column Name	Data Type	Constraints	Description
id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for the admin.
name	VARCHAR(255)	NOT NULL	Name of the administrator.
email	VARCHAR(255)	NOT NULL	Admin email address.
password	VARCHAR(255)	NOT NULL	Hashed login password.
code	MEDIUMINT(50)	NOT NULL	Verification code.
status	TEXT	NOT NULL	Account status.

5.2.4. Table: Contact

Stores queries or feedback submitted by visitors via the "Contact Us" form.

Column Name	Data Type	Constraints	Description
fname	VARCHAR(20)	NOT NULL	First Name.
lname	VARCHAR(20)	NOT NULL	Last Name.
contactEmail	VARCHAR(20)	NOT NULL	Email for reply.
contactPhone	VARCHAR(10)	NOT NULL	Contact number.
comment	VARCHAR(2000)	NOT NULL	The query or message body.

5.2.5. Table: Admin login (Legacy/Super Admin)

A simpler table used for initial or super-admin access.

Column Name	Data Type	Constraints	Description
Id	INT(11)	PRIMARY KEY, AUTO_INCREMENT	Unique ID.
username	VARCHAR(255)	NOT NULL, Default 'admin'	Username for basic login.
password	VARCHAR(255)	NOT NULL, Default 'admin'	Password for basic login.

5.3. Data Integrity and Normalization

- **Primary Keys**: Every table (except contact) utilizes an INT Primary Key (id) with AUTO_INCREMENT to ensure unique record identification.
- **Data Types**: VARCHAR is optimized for variable-length text (names, emails) to save storage, while TEXT is reserved for status flags. INT is used for efficient indexing.
- **Relationships**: The garbageinfo table is logically linked to usertable via the email field, allowing the system to track history for specific users.

6 RESULT , DISCUSSION AND CONCLUSION

6.1. RESULT

6.2. CONCLUSION

6.1. Result

Fig 6.1.1: Website main page

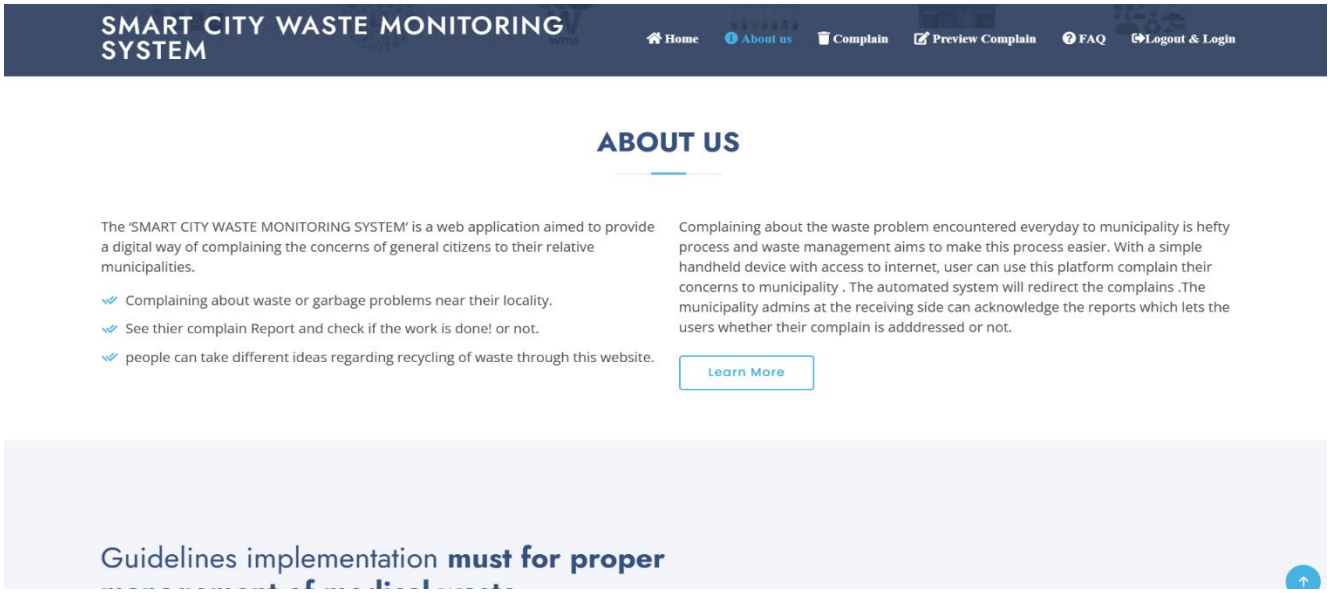

Fig 6.1.2: About us

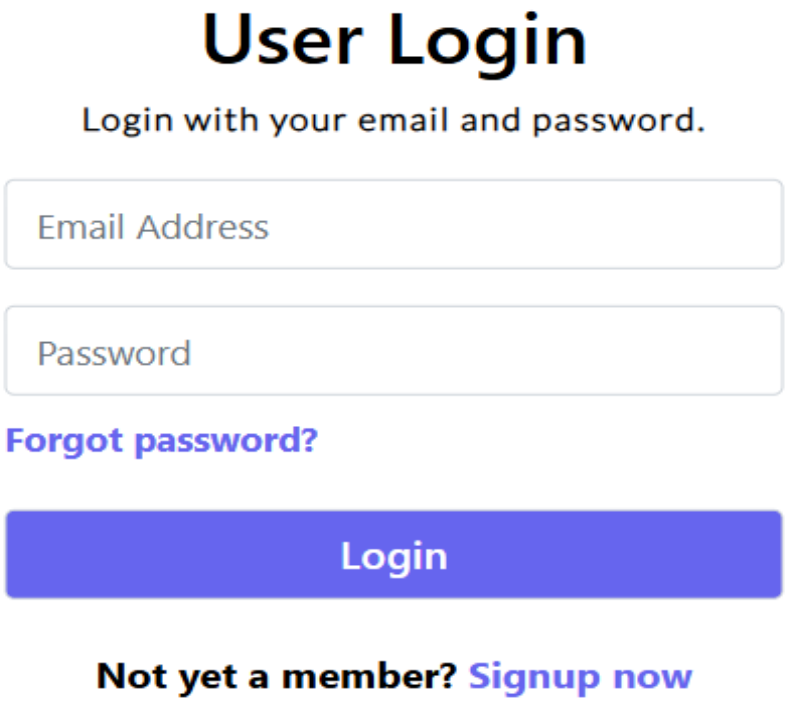

Fig 6.1.3: User login

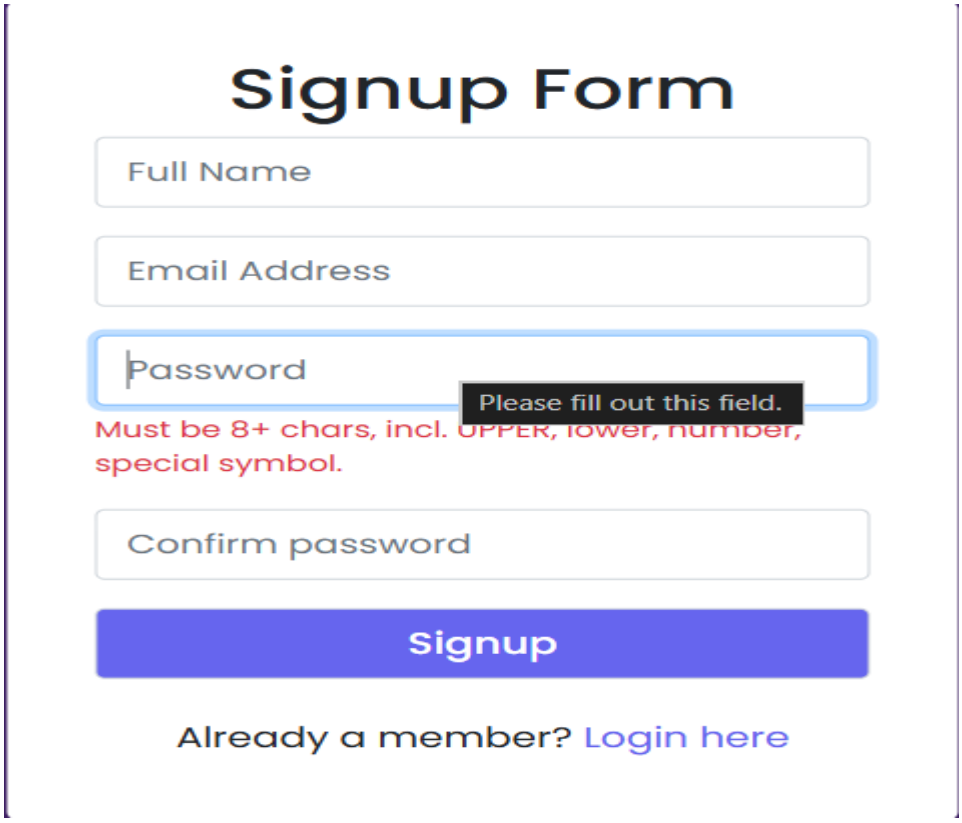

Fig 6.1.4: Signup form

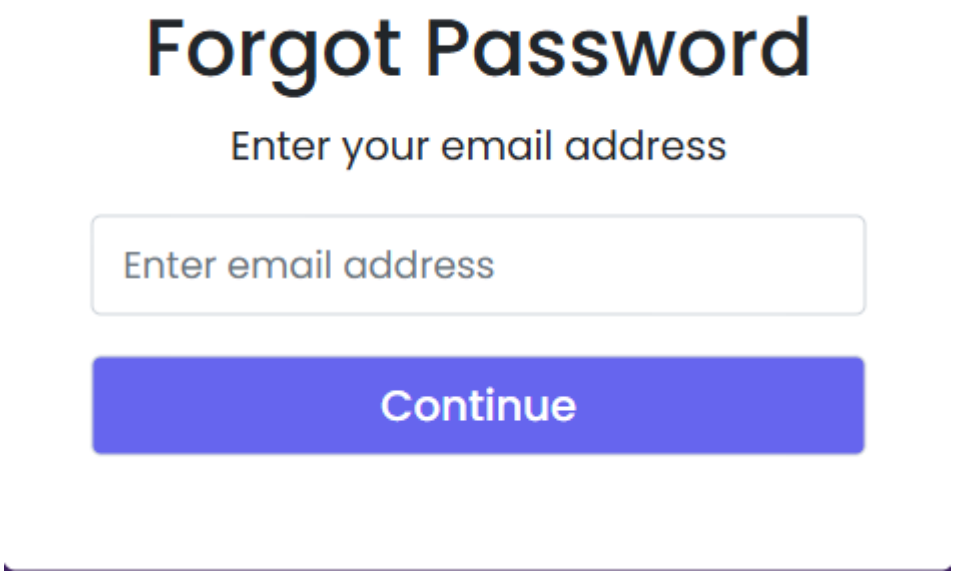

Fig 6.1.5: Forgot password

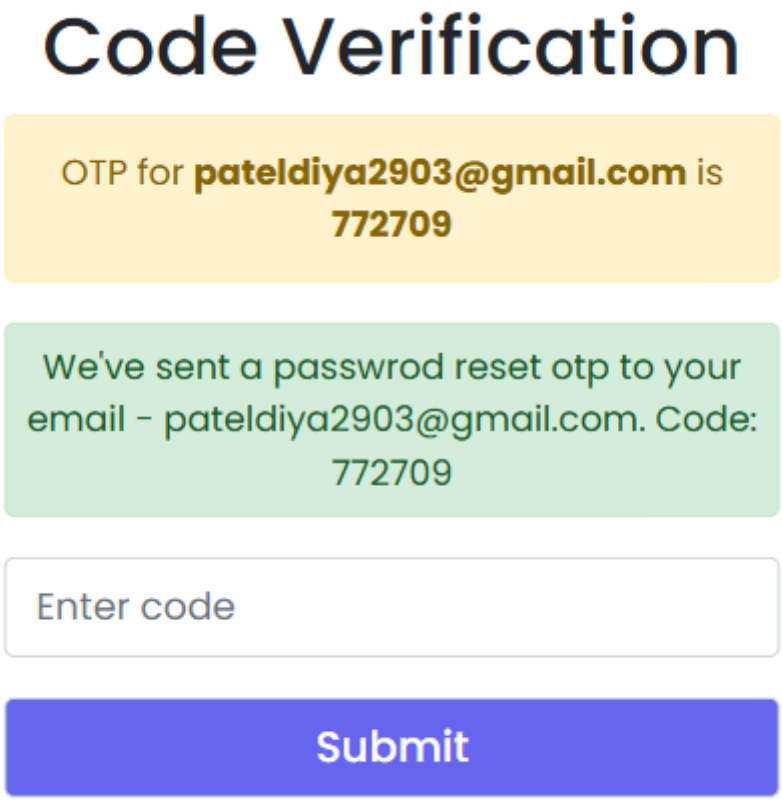

Fig 6.1.6: Code verification

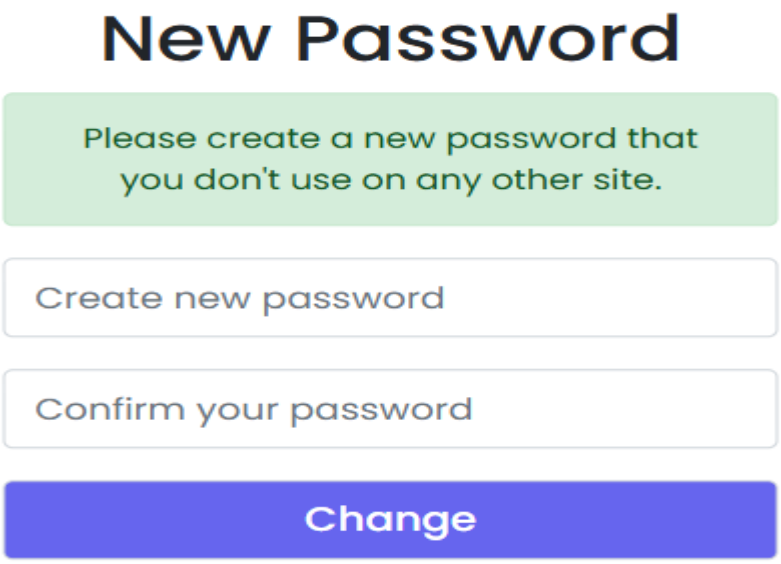

Fig 6.1.7: New password

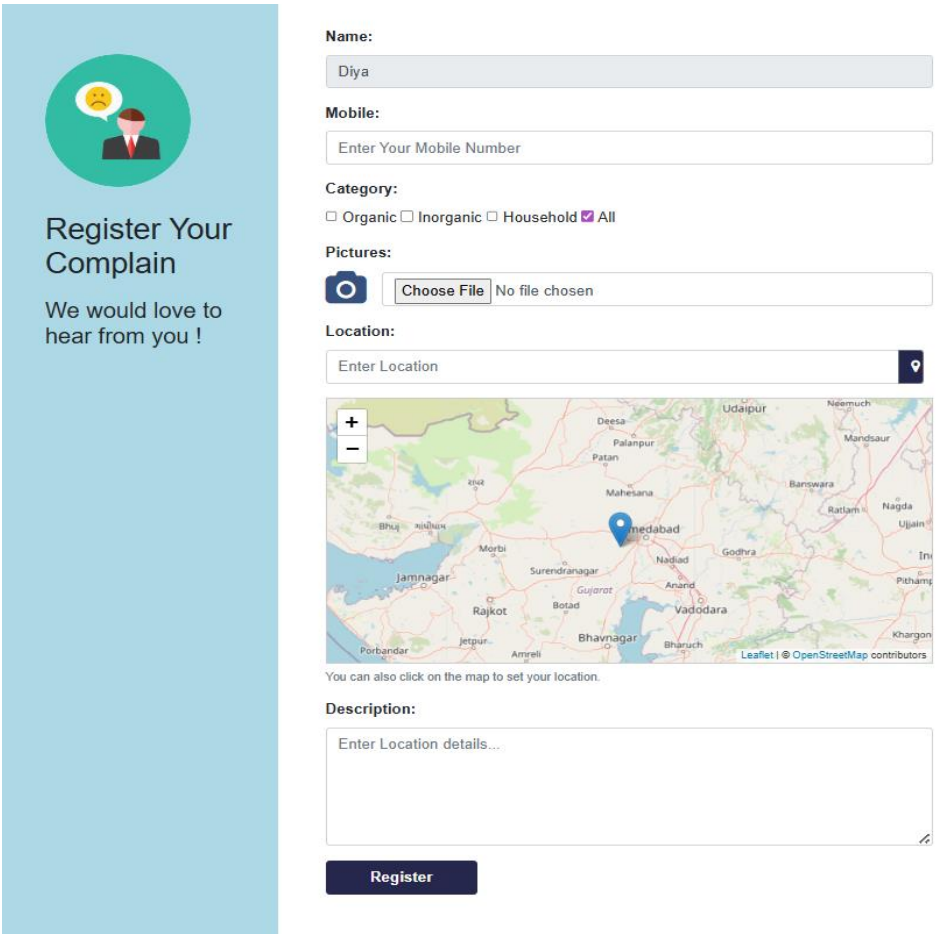

Fig 6.1.8: Complain registration

YOUR COMPLAIN HISTORY

Date	Name	Mobile	Email	Type of Waste	Location Details	Images	Status	Operations
8:15am , Monday 26th January 2026	Diya	9106664008	pateldiya2903@gmail.com	organic,Household,	**Loc:** ahmedabad naroda **Desc:** near civil hospital		Approved	Delete Edit
9:15am , Thursday 29th January 2026	Diya	0133466451	pateldiya2903@gmail.com	mixed,	**Loc:** valsad 396001 **Desc:** ghadoi vadi faliya valsad		Pending	Delete Edit
9:29am , Thursday 29th January 2026	Diya	0123456789	pateldiya2903@gmail.com	mixed,	**Loc:** amreli **Desc:** near hospital		Pending	Delete Edit

Fig 6.1.9: Complain history(Preview Complain)

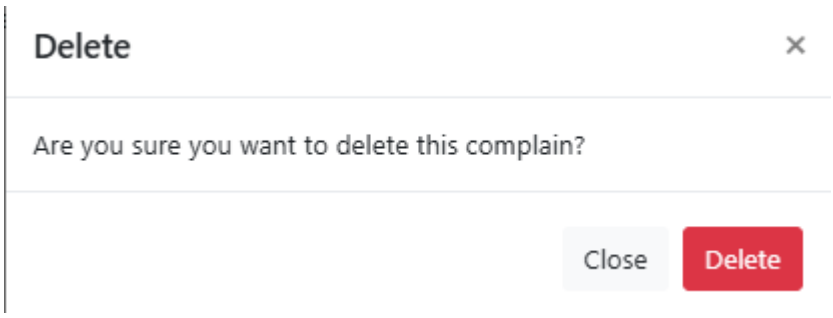

Fig 6.1.10: Delete Complain

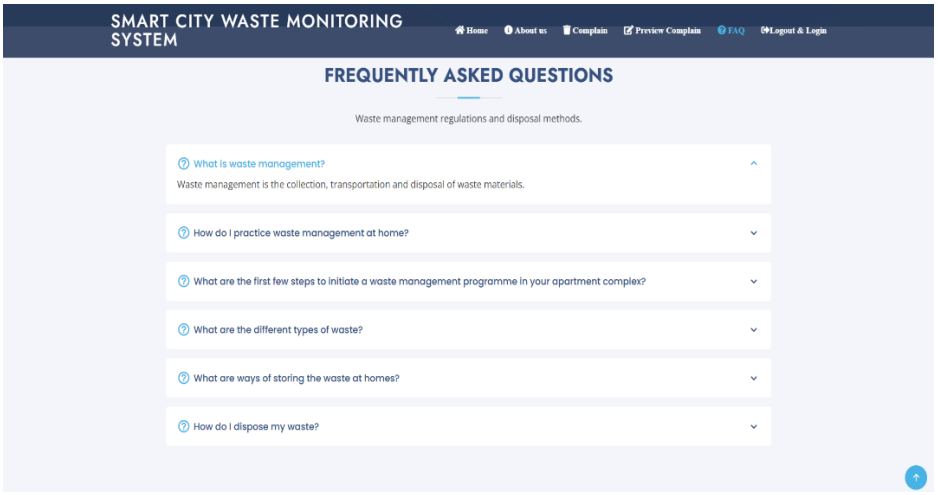

Fig 6.1.11: Frequently asked questions

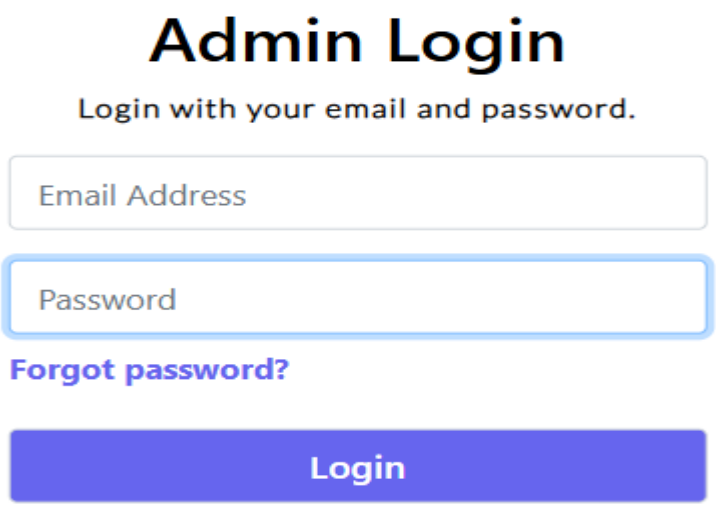

Fig 6.1.12: Admin login

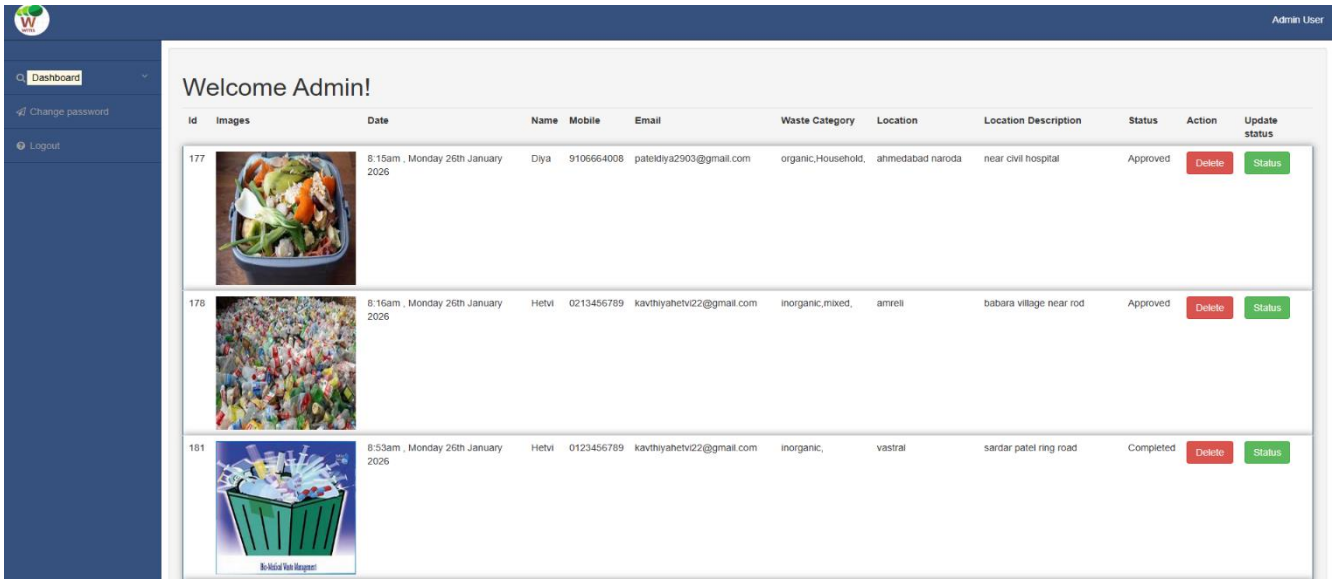

Fig 6.1.13: Admin Dashboard

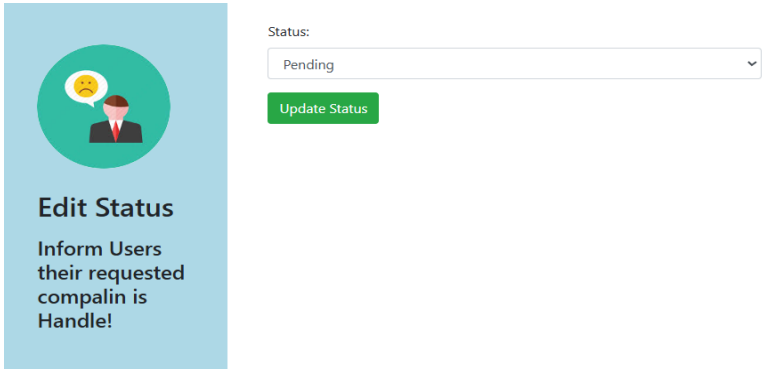

Fig 6.1.14: Edit status

6.2. Conclusion

The Smart City Waste Monitoring System bridges the gap between citizens and municipal authorities by providing a digital platform for real-time waste reporting. By replacing inefficient manual processes with a centralized reporting dashboard, the system enhances accountability and speeds up waste collection. Ultimately, it promotes a cleaner environment and public health awareness through active citizen participation and data-driven governance.

7 BIBLIOGRAPHY

7.1. REFERENCES

7.1. References

This section lists the academic papers, technical documentation, and online resources referenced during the development and research of the **Smart City Waste Monitoring System**.

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7.1.2. Technical Documentation

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- **Bootstrap 5 Documentation**. [Online].
  Available: https://getbootstrap.com/docs/5.0/getting-started/introduction/. [Accessed: Jan 2024]. (Used for the responsive design of index.html).
- **Google Maps Platform Documentation.** [Online].
  Available: https://developers.google.com/maps/documentation. (Relevant for future GIS integration).

7.1.3. Web Resources & Libraries

- **AOS - Animate On Scroll Library**. [Online].
  Available: https://michalsnik.github.io/aos/. (Used in assets/vendor/aos for UI animations).
- **Swiper - Mobile Touch Slider**. [Online].
  Available: https://swiperjs.com/. (Used in assets/vendor/swiper for the testimonial section).
- **PHPCodex - PHP Mailer Guide**. [Online].

Available: https://phpcodex.com/send-email-using-phpmailer/. (Reference implementation for phpGmailSMTP).

7.1.4. Similar Systems (Case Studies)

- **Bigbelly Smart Waste & Recycling System**. [Online].
  Available: https://bigbelly.com/. (Industry standard for smart bins).
- **Ecube Labs - Smart Waste Management**. [Online].
- Available: https://www.ecubelabs.com/.